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ADAN 8888 - Applied Analytics Project - Spring 2026

Module 10 Assignment - Data-Centric AI

HelpHerInvest: A machine learning-based investment recommendation system that aligns user interests and risk preferences with benchmark stock evaluation.

Three Data-Centric AI Approaches

Cross-sectionally Ranked Dataset

We created a cross-sectionally ranked dataset by converting key predictors, such as momentum, relative SPY performance, volatility, drawdown, and percent above the 200-day moving average, into within-date percentile ranks. This improved the data by making features more comparable across stocks on the same date and by helping the model learn relative strength patterns instead of being affected by differences in raw scale. This is especially useful in a stock-selection setting where relative ranking often matters more than absolute values.

Skinny Dataset

We built a skinny dataset that kept only a small subset of 6-month features: mom_6m, rel_6m_spy, vol_6m, and drawdown_6m. This simplified the input data and reduced potential noise from less informative variables. By narrowing the dataset to a focused group of features, we tested whether a cleaner and lower-dimensional representation of the data could improve model performance and generalization.

Additional Engineering Features of Existing Data

We developed an engineered-features dataset by adding new variables derived from the original columns. These included mom_spread_short_long, mom_acceleration, vol_ratio, mom_vol_adj, recovery_signal, and trend_strength. These new features were intended to capture more meaningful relationships in the data, such as changes in momentum, volatility-adjusted return behavior, recovery from drawdowns, and overall trend quality. This helped enrich the dataset with signals that may be more predictive than the raw features alone.

Error Analysis and Data Improvements

Our error analysis focused on the validation classification reports and confusion matrices for the three LightGBM models trained on different versions of the dataset. The cross-sectionally ranked dataset produced the best overall validation results, with accuracy = 0.52, macro F1 = 0.5175, and ROC-AUC = 0.5234, outperforming the skinny-feature dataset (F1 = 0.4991, ROC-AUC = 0.5007) and the engineered-feature dataset (F1 = 0.5091, ROC-AUC = 0.5152). The confusion matrix for the ranked model also showed the most balanced performance across

both classes, with 13,127 true negatives, 11,703 false positives, 11,180 false negatives, and 11,478 true positives.

In contrast, the skinny dataset performed almost like random guessing, suggesting that reducing the data to only four 6-month features removed too much useful information.
The engineered-feature dataset performed slightly better than the skinny version, but it showed weaker recall for class 1, indicating that it still struggled to identify positive excess-return cases consistently.

As a result of this error analysis, we made data-centric improvements in three main ways. First, we used cross-sectional ranking to reduce scale differences and improve comparability across stocks on the same date, which turned out to be the most effective change. Second, we tested feature reduction through the skinny dataset to evaluate whether a simpler dataset would reduce noise, although the results showed that too much simplification hurt predictive power. Third, we added engineered features such as momentum spreads, acceleration, volatility ratios, recovery signals, and trend strength to capture more meaningful patterns from the raw inputs. We also improved data quality by dropping missing rows before modeling, reducing the dataset from 302,024 to 274,511 rows. Overall, the error analysis showed that improving feature representation mattered more than simply shrinking the feature set.

Retraining the LightGBM Model

For this step, we refit and retrained the Week 9 selected model (LightGBM 2) using the updated training datasets created in Week 10. Rather than changing the algorithm, we kept the same model family and trained three LightGBM versions on the revised data representations: the cross-sectionally ranked dataset, the skinny 6-month feature dataset, and the engineered-features dataset. The main difference in performance came from the data updates rather than model tuning, as all three models used the same configuration:

- n_estimators=1000
- learning_rate=0.01
- max_depth=12
- num_leaves=31
- class_weight="balanced"

The retraining was done using a time-based 70/15/15 train-validation-test split, which preserved the chronological structure of the financial data. Among the retrained versions, the LightGBM model trained on the cross-sectionally ranked dataset performed best on validation, showing that the data improvements were more effective than simply reducing features or adding engineered features alone.

Performance of the Datasets

Using the updated validation dataset, we compared the retrained Week 9 model family, LightGBM, across the three Week 10 data-centric dataset versions. The ranked-feature version clearly produced the strongest validation metrics among the retrained models.

1. The **cross-sectionally ranked** LightGBM (Model 1) achieved the best validation performance with accuracy = 0.52, macro F1 = 0.5175, and ROC-AUC = 0.5234.

```
LightGBM Model 1:
Classification Report:

              precision    recall  f1-score   support

           0       0.54      0.53      0.53     24830
           1       0.50      0.51      0.50     22658

    accuracy                           0.52     47488
   macro avg       0.52      0.52      0.52     47488
weighted avg       0.52      0.52      0.52     47488

F1: 0.5175
ROC-AUC: 0.5234
```

2. The **skinny-feature** LightGBM (Model 2) performed worst, with accuracy = 0.50, macro F1 = 0.4991, and ROC-AUC = 0.5007.

```
LightGBM Model 2:
Classification Report:

              precision    recall  f1-score   support

           0       0.53      0.49      0.51     27877
           1       0.47      0.51      0.49     24827

    accuracy                           0.50     52704
   macro avg       0.50      0.50      0.50     52704
weighted avg       0.50      0.50      0.50     52704

F1: 0.4991
ROC-AUC: 0.5007
```

3. The **engineered-features** LightGBM (Model 3) was in between, with accuracy = 0.51, macro F1 = 0.5091, and ROC-AUC = 0.5152.

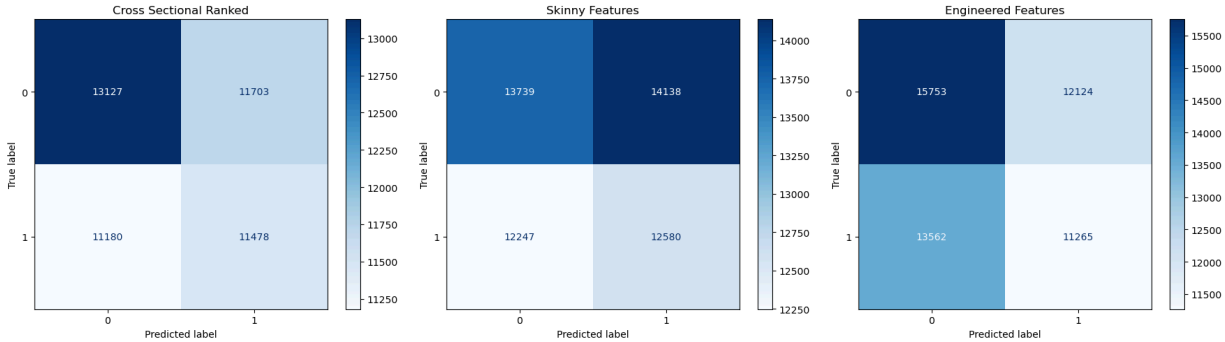
```
LightGBM Model 3:
Classification Report:

              precision    recall  f1-score   support

           0       0.54      0.57      0.55     27877
           1       0.48      0.45      0.47     24827

    accuracy                           0.51     52704
   macro avg       0.51      0.51      0.51     52704
weighted avg       0.51      0.51      0.51     52704

F1: 0.5091
ROC-AUC: 0.5152
```



The cross-sectionally ranked LightGBM performed better because ranking the features by date improved comparability across stocks and reduced the effect of raw-scale differences. In a cross-sectional stock prediction problem, relative position is often more informative than absolute feature magnitude, and the validation results support that. The skinny dataset likely removed too much useful information, while the engineered-features dataset added potentially helpful signals but still did not outperform the ranked representation.
Therefore, we selected the LightGBM model trained on the cross-sectionally ranked dataset as our final model to deploy, because it delivered the best balance of validation accuracy, F1 score, and ROC-AUC among all Week 10 retrained candidates.

Final Takeaways

The selected model's validation and test results are very close, which suggests the model generalizes reasonably well to unseen data. There is no strong sign of severe overfitting based on the reported outputs.
Overall model performance is still only slightly better than random guessing, especially from the ROC-AUC values just above 0.50. This means the data improvements helped, but the prediction problem remains difficult. The model is able to extract some useful signal, but the signal is weak, so further gains will likely require additional data improvements, better labeling strategies, or stronger predictive features rather than simply retraining the same setup.